

Problem Sets After Test 2

*Only turn in problems that are **not** bracketed.* Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

notation	meaning
unbracketed	assigned problem – turn these in for grading
()	helper/warm-up problem
[]	additional problems (you are responsible for content, but don't turn them in)
{ }	covers optional material

Due	Task	Source	Problems
Wed 4/7	PS 13	Rosen 13.3	(11) testing strings 12 testing strings 16–17 what language? 25–26 create DFA
Fri 4/9	PS 14	Rosen 13.3	19–20 what language? [32] create DFA 34 create DFA 36 create NFA 45–46 what language? [47] what language? 55–56 create DFA, NFA Note: Please do 55 and 56 in this order: 55a, 56a, 55b, 56b, 55c, 56c.
Wed 4/14	PS 15	Rosen 13.3 Rosen 13.4	52–54 NFA to DFA [1] 2abde describe in words [3] 4 does it belong? 6 regex [7] regex
Mon 4/19	PS 16	Rosen 13.4	14abc grammar to NFA 15–17 NFA to grammar
Wed 4/21	PS 17	Rosen 13.4 DFA/NFA → regex	12ab regex to NFA 13a regex to NFA Convert the following DFAs or NFAs into equivalent regular expressions using our algorithm for doing so. Show the intermediate steps: 13.3.43 13.3.45 (Note: these are not the instructions in the book!)
Fri 4/23	PS 18	Rosen 13.4	23, 25 pumping lemma Be sure to write the up carefully.
Mon 4/26	PS 19	Rosen 13.5	2 trace TM 5 trace TM [1] trace TM [3–4] trace TM
Wed 4/28	PS 20	Rosen 13.5	6 design TM 10 design TM [7–12] design TM For problems 6 and 10, describe your Turing machine using the syntax of one of our two Turing machine simulators. (You can use a text editor or copy and paste from the online simulator into a file to save your work.)

Due	Task	Source	Problems
never	Review	Rosen 13.RQ Rosen 13.SE	[1–2] [3] [4] [5a] [6a–12] [15] [17] [19] [3] [7] [8] [9] [19–21] [23–25] [26–27]
never	Review	Rosen 6.RQ Rosen 6.SE Rosen 7.RQ Rosen 7.SE Rosen 10.SE Rosen 10.RQ	[1–2] counting [5] counting [6–7] PHP [13] collections [1–7] counting [11] fortune cookies [36] round table [37] advisors [38] apples [42] license plate [1–6] probability basics [9] E is linear [1–5] probability [7a] expected value [17] cards in a box [23–24] [35] aces [3–5] isomorphic? [29abcf] invariants [1] types of graph [2] models [3–5] degree [6–7] graph examples [8] bipartite [9] graph representation [10] isomorphism [11] connected [12] checking isomorphism [13] Euler paths [14a] Hamilton paths [17] planar graph [18] Euler's formula [19] Kuratowski's Theorem